

Minimal Cover / Canonical Cover

Also known as Irreducible Cover • Step-by-Step Algorithm & Solved Examples

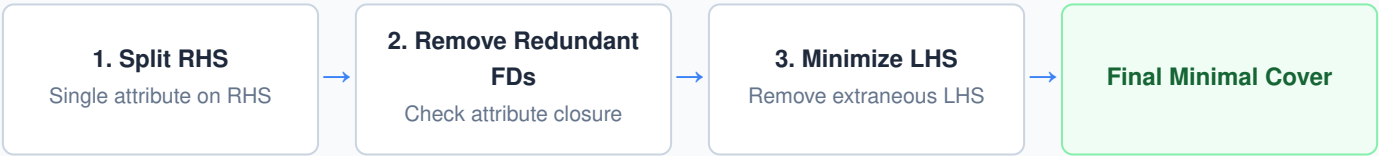
A **Minimal Cover** (or **Canonical Cover** / **Irreducible Cover**) of a set of functional dependencies (FDs) is a minimal equivalent set that retains all the dependency information of the original set without any redundancy or extraneous attributes.

★ **Exam-Ready Definition**

A minimal cover is an equivalent set of functional dependencies in which **every FD has a single attribute on the RHS**, **no FD is redundant**, and **no attribute on the LHS of any FD is extraneous**.

3 Steps to Find Minimal Cover

Memory Rule: "Split Right, Remove Redundancy, Minimize Left" ($R \rightarrow \textit{Redundant} \rightarrow L$)



Given Example Problem

Consider the set of functional dependencies F :

$F = \{$
 $A \rightarrow B,$
 $C \rightarrow B,$
 $D \rightarrow ABC,$
 $AC \rightarrow D$
 $\}$

Step 1: Make RHS Single Attribute ★

Every functional dependency must have exactly one attribute on the Right-Hand Side (RHS).

FD	RHS Status	Action / Result
$A \rightarrow B$	✓ Single	Keep as is
$C \rightarrow B$	✓ Single	Keep as is
$D \rightarrow ABC$	✗ Multiple	Decompose into $D \rightarrow A, D \rightarrow B, D \rightarrow C$
$AC \rightarrow D$	✓ Single	Keep as is (LHS stays intact)

⚠ Golden Rule & Exam Trap

RHS can be decomposed; LHS cannot be decomposed directly.

$D \rightarrow ABC$ becomes $D \rightarrow A, D \rightarrow B, D \rightarrow C$.

However, $AC \rightarrow D$ **cannot** be changed into $A \rightarrow D$ and $C \rightarrow D$ because A and C may be required together to determine D .

Set after Step 1:

$F = \{ A \rightarrow B, C \rightarrow B, D \rightarrow A, D \rightarrow B, D \rightarrow C, AC \rightarrow D \}$

Step 2: Remove Redundant Functional Dependencies

To test if $X \rightarrow Y$ is redundant:

1. Temporarily remove $X \rightarrow Y$ from the set.
2. Calculate X^+ (attribute closure of X) using the remaining FDs.
3. If $Y \in X^+$, then $X \rightarrow Y$ is **redundant** (can be derived) and should be removed. Otherwise, keep it.

1. Check $A \rightarrow B$

Remove $A \rightarrow B$. Remaining FDs: $\{C \rightarrow B, D \rightarrow A, D \rightarrow B, D \rightarrow C, AC \rightarrow D\}$.

Calculate closure: $A^+ = \{A\}$. Since $B \notin A^+ \rightarrow$ **✓ Keep $A \rightarrow B$**

2. Check $C \rightarrow B$

Remove $C \rightarrow B$. Remaining FDs: $\{A \rightarrow B, D \rightarrow A, D \rightarrow B, D \rightarrow C, AC \rightarrow D\}$.

Calculate closure: $C^+ = \{C\}$. Since $B \notin C^+ \rightarrow$ **✓ Keep $C \rightarrow B$**

3. Check $D \rightarrow A$

Remove $D \rightarrow A$. Remaining FDs: $\{A \rightarrow B, C \rightarrow B, D \rightarrow B, D \rightarrow C, AC \rightarrow D\}$.

Calculate closure: $D^+ = \{D, B, C\}$. Since $A \notin D^+ \rightarrow$ **✓ Keep $D \rightarrow A$**

4. Check $D \rightarrow B$

Remove $D \rightarrow B$. Remaining FDs: $\{A \rightarrow B, C \rightarrow B, D \rightarrow A, D \rightarrow C, AC \rightarrow D\}$.

Calculate closure of D :

- Start: $D^+ = \{D\}$
- Using $D \rightarrow A$: $D^+ = \{D, A\}$
- Using $A \rightarrow B$: $D^+ = \{D, A, B\}$

Since $B \in D^+$ via transitive derivation ($D \rightarrow A$ and $A \rightarrow B$) $\rightarrow$ **✗ Redundant — REMOVE $D \rightarrow B$**

5. Check $D \rightarrow C$

Remove $D \rightarrow C$. Remaining FDs: $\{A \rightarrow B, C \rightarrow B, D \rightarrow A, AC \rightarrow D\}$.

Calculate closure: $D^+ = \{D, A, B\}$. Since $C \notin D^+ \rightarrow$  **Keep $D \rightarrow C$**

6. Check $AC \rightarrow D$

Remove $AC \rightarrow D$. Remaining FDs: $\{A \rightarrow B, C \rightarrow B, D \rightarrow A, D \rightarrow C\}$.

Calculate closure: $(AC)^+ = \{A, C, B\}$. Since $D \notin (AC)^+ \rightarrow$  **Keep $AC \rightarrow D$**

Set after Step 2:

$F = \{ A \rightarrow B, C \rightarrow B, D \rightarrow A, D \rightarrow C, AC \rightarrow D \}$

Step 3: Minimize the LHS (Remove Extraneous Attributes)

Check whether any multi-attribute LHS contains an extraneous attribute.

In our current set, only $AC \rightarrow D$ has multiple attributes on the LHS.

Test 1: Can we remove A? (Is A extraneous?)

- If A is removed, $AC \rightarrow D$ becomes $C \rightarrow D$.
- Calculate C^+ using current set: $C^+ = \{C, B\}$.
- Since $D \notin C^+$, attribute A cannot be removed.

Test 2: Can we remove C? (Is C extraneous?)

- If C is removed, $AC \rightarrow D$ becomes $A \rightarrow D$.
- Calculate A^+ using current set: $A^+ = \{A, B\}$.
- Since $D \notin A^+$, attribute C cannot be removed.

Therefore, $AC \rightarrow D$ must remain as it is.

Final Minimal Cover

The safest and standard exam representation (with single attribute on RHS):

```
Fmin = {  
  A → B,  
  C → B,  
  D → A,  
  D → C,  
  AC → D  
}
```

Note: You may also write $D \rightarrow A$ and $D \rightarrow C$ combined as $D \rightarrow AC$:

$$F_{\min} = \{ A \rightarrow B, C \rightarrow B, D \rightarrow AC, AC \rightarrow D \}$$

? Why Can't We Write $D \rightarrow ABC$ in the Final Answer?

While combining same LHS attributes (e.g., $D \rightarrow A$ and $D \rightarrow C$ into $D \rightarrow AC$) is mathematically equivalent, formal academic definitions of minimal cover require **strictly a single attribute on the RHS**. Unless specified by MCQ options, stick to single-RHS format.

Important Exam Traps ⚠

Trap Type	Common Mistake	Correct Practice
Trap 1: Splitting LHS	Splitting $AC \rightarrow D$ into $A \rightarrow D$ and $C \rightarrow D$.	Never split LHS directly! LHS attributes can only be tested/removed via Step 3 closure.
Trap 2: Redundancy Test	Calculating closure without removing the target FD first.	Always temporarily remove $X \rightarrow Y$ before computing X^+ .
Trap 3: Transitivity	Missing indirect derivations like $D \rightarrow A$ and $A \rightarrow B$ deriving $D \rightarrow B$.	Always perform full closure calculations to catch transitive dependencies.

Minimal Cover vs. Candidate Key

Feature	Minimal Cover	Candidate Key
Primary Objective	Simplifies set of Functional Dependencies	Uniquely identifies tuples in a relation
Redundancy Removal	Removes redundant FDs and extraneous attributes	Minimal superkey (no redundant attributes)
Core Tool	Uses FD attribute closure (X^+)	Uses FD attribute closure (X^+)
Application	Database Normalization (3NF/BCNF decomposition)	Primary key selection, schema design

Minimal Cover = Equivalent FDs + Single RHS + No Redundant FD + No Extraneous LHS Attribute